

Explainer: URLs

Duke OLLI Digital Explorers SIG

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Definition of URL

- A URL is the address of a page
- “URL” is an acronym for **U**niversal **R**esource **L**ocator

Merriam-Webster

: the address of a resource (such as a document or website) on the Internet that consists of a communications protocol followed by the name or address of a computer on the network and that often includes additional locating information (such as directory and file names)

| our site's URL is <http://www.Merriam-Webster.com>

→ called also uniform resource locator, universal resource loca

American Heritage Dictionary

noun

1. An Internet address (for example, <http://hmhbooks.com/eref/>), usually consisting of the access protocol (http), the domain name (hmhbooks.com), and optionally the path to a file or resource residing on the server where the domain name resides (eref).
2. The address of a web page on the world wide web.

initialism

1. A Uniform Resource Locator: the address of a web page, ftp site, audio stream or other Internet resource.

Anatomy of a URL

protocol://hostname/resource?parameters

Uniform Resource Locator

Protocol

Resource

<https://olli-shamlin.github.io/spring-2024/syllabus.html>

Host Name
or Domain Name

URL Protocol

***protocol**://hostname/resource?parameters*

- Usually HTTP or HTTPS
- The site determines whether HTTP or HTTPS is used
When you enter a URL into the Address Bar, you don't have to type the protocol
- HTTP is an acronym for
Hyper**T**ext **T**ransport **P**rotocol
- The “S” in “HTTPS” stands for “secure”
When HTTPS is used...
 - Communication between the browser and the site is **encrypted**
 - The browser can “prove” the site's identity using a **certificate** sent to the browser by the site

URL Hostname

protocol://hostname/resource?parameters

- The hostname provides the address of the site
- The value found here is typically a **domain name**
 - Think of “domain name” as a synonym for “site name”
- Domain names are “words” separated by periods; for example...
 - www.amazon.com
 - en.wikipedia.org
 - courses.learnmore.duke.edu
- Note: domain names are used in email addresses!
E.g., “gmail.com” is the domain in the email address “dukeollitotw@gmail.com”

URL Resource

*protocol://hostname/**resource?**parameters*

- The resource is the address of the page
- The value found here is usually a list of words separated by forward slashes ('/')
- Sometimes the value may also contain

- **A file extension**

`https://olli-shamlin.github.io/spring-2024/syllabus.html`

- **An anchor point**

`https://en.wikipedia.org/wiki/IP_address#Subnetworks`

URL Paramters

protocol://hostname/resource?parameters

- Parameters are pieces of information your browser sends to the site in addition to the resource
- When parameters are needed is determined by the page currently loaded in the View Pane
- Your browser generates parameters part of a URL when needed
- Don't be surprised if/when the parameters value looks like "gibberish"
Your browser and the site you are visiting know how to "read" them

Example of a URL with parameters

[https://www.google.com/search?](https://www.google.com/search?q=what+is+a+url&sca_esv=ef32433310457e91&sca_upv=1&ei=FUwiZsuiFcOs5NoP_7CMyAk&ved=0ahUKEwiLgcOrjc6FAxVDFIkFHX8YA5kQ4dUDCBA&uact=5&oq=what+is+a+url&gs_lp=Egxnd3Mtd2l6LXNlcnAiDXdoYXQgaXMgYSB1cmwyCBAA)

q=what+is+a+url&sca_esv=ef32433310457e91&sca_upv=1&ei=FUwiZsuiFcOs5NoP_7CMyAk&ved=0ahUKEwiLgcOrjc6FAxVDFIkFHX8YA5kQ4dUDCBA&uact=5&oq=what+is+a+url&gs_lp=Egxnd3Mtd2l6LXNlcnAiDXdoYXQgaXMgYSB1cmwyCBAA
GIAEGLEDmgUQABiABDIFEAAyGAQyBRAAGIAEMgUQABiABDIFEAAyGAQyBRA
AGIAEMgUQABiABDIFEAAyGAQyBRAAGIAESPAeUMYNWJgccAR4AZABAJgBZK
ABuAaqAQQxMi4xuAEDyAEA-
AEBmAIRoAKOB8ICChAAGLADGNYEGEfCAg0QABiABBiwAxhDGloFwgITEC4Yg
AQYsAMYQxjIAxiKBdgBAcICEBAAGIAEGLEDGEMYgwEYigXCAGsQLhiABBixAxi
DAcICCxAAGIAEGLEDGIMBwgIREC4YgAQYsQMY0QMYgwEYxwHCAg4QLhiAB
BixAxjRAxjHAcICChAAGIAEGEMYigXCAG4QLhiABBixAxiDARiKBcICDhAAGIAEGL
EDGIMBGloFwgINEAAyGAQYsQMYQxiKBcICCBAAGIAEGMkDwgILEAAyGAQYkg
MYigXCAGsQLhiABBIRAhIKBcICBBAAGAPCAiMQLhiABBIRAhIKBRiXBRjcBBjeBB
jgBBj0AxjxAxj1A9gBApgDAIgGAZAGDLoGBAgBGAI6BgYIAhABGBSSBwQxNi4xo
AeVZg&sclient=gws-wiz-serp